

Peter Shr

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Objective: Electrical engineering student at UCSD experienced in PCB design and power electronics interested in semiconductor devices

Skills

- **Programming:** C, Python, SQL, HTML5, CSS, JS | **Hardware Description:** Verilog (basic digital logic design)
- **PCB Design:** EasyEDA, KiCad, Altium | **CAD:** Onshape, SolidWorks
- **Electrical:** Circuit analysis, phasor analysis, Fourier series & transforms | **Tools:** Multimeter, soldering, Arduino

Education

University of California, San Diego – La Jolla, CA

Sep 2025 - Present

- Bachelor of Science in Electrical Engineering, Expected 2029 - GPA: 4.0 | Junior Standing
- Relevant coursework: Analog Design, Digital Design, Circuits and Systems, Multivariable Calculus, Differential Equations

American High School – Fremont, CA

Aug 2021 - Jun 2025

- Honor Graduate (Top 10 in graduating class) - GPA: 4.0 unweighted

Work Experience

Erogbogbo Research Lab, SJSU – Undergraduate Research Assistant

Jun 2026 – Present

- Expanding ObservarXiv platform by developing a web-based instrument submission module supporting FTIR characterization, including binary Perkin Elmer .SP file parsing and automated SOP field extraction
- Working toward standardized protocol documentation for semiconductor and failure analysis instrumentation to improve measurement reproducibility

Mosi Audio – Electrical Engineering Intern

May 2026 - Present

- Designing battery power capability into the Mosi Nova using the MCP73871 power path management IC to switch between USB-C input and battery supply
- Implementing the CME WIDI Core Bluetooth MIDI module to enable wireless connectivity to the Mosi Nova's custom PCB

Center for Community Energy – Electrical Engineering Intern

Mar 2026 - Present

- Analyzed multiple solar PV system electrical designs in HelioScope to verify NEC 120% busbar rule compliance and identify compatible components and quantities for power control systems
- Designed a 685 kW solar carport system in HelioScope for a Viasat parking structure, optimizing module layout, inverter selection, tilt angle, and AC voltage drop while ensuring NEC 120% busbar compliance

Triton Robotics – Electronics Subteam

Jan 2026 - Present

- Designed a four-layer breakout PCB on EasyEDA to connect NVIDIA Jetson with BNO055 IMU, implementing I2C communication with pull-up resistors
- Designed a custom ISM330DLC IMU PCB from scratch in EasyEDA, removing SPI support, integrating XH connectors for I2C, and implementing noise reduction techniques including strategic decoupling capacitor placement, controlled trace widths, and a ground plane with minimal penetration
- Soldered, crimped, and assembled PCB components and wiring harness to reduce manufacturing cost

Human Powered Submarine – Electronics Subteam

Sep 2025 - Present

- Designed, 3D-printed, and assembled an analog Hall-effect joystick for pilot control using SolidWorks and makerspace tools
- Developing Arduino code to read joystick direction based on magnet position relative to Hall-effect sensors

Projects

COSMOS – UCSC Cluster 10: Semiconductor Materials and Device Engineering

Jul 2024 - Aug 2024

- Studied semiconductor device physics including PN junctions, MOSFETs, BJTs, and optoelectronic systems
- Investigated and produced a research poster on carbon nanotube FET architectures and their scalability limits in nanoelectronics

Volunteering

Build Your Dreams – Co-Founder & Vice President

Dec 2023 - Jun 2025

- Organized and led hands-on robotics lessons to teach students CAD, breadboarding, programming in workshops set in schools and libraries in the Bay Area
- Cooperated with local robotics teams to showcase their robots to expand interest in STEM

Awards

- Academic: AP Scholar Award, LAMVCF Science for Good Scholarship, AHS Honor Graduate, Commended Scholar
- Others: 2x North Coast Section Badminton, CIF Scholar-Athlete Award, Norma Teagarden Piano Scholarship